

BRIEFING · AUGUST 2026

Artificial Intelligence in *UK Energy Retail*

What the evidence actually says, what Ofgem expects, and what can safely be done now.

Neil Catton

Independent Technology Strategist · Technology, Ethics & Human Purpose

NEILCATTON.COM

Prepared as a market briefing. Every figure is traced to a regulator, a trade body, an ombudsman, published legislation, or an operator speaking on its own record.

A note on the numbers

Almost every published statistic about AI in energy retail is produced by a company selling AI into energy retail.

A briefing built on those numbers is indistinguishable from the vendor material a supplier's executive team already ignores, so this one is not built on them. The rule applied throughout is narrow and it is applied without exception: if a figure cannot be traced to Ofgem, the Energy Ombudsman, published legislation, a government consultation outcome, or an operator speaking on its own record, it does not appear here. Where a figure was found and rejected, the rejection is recorded below rather than quietly absorbed.

This matters more in energy retail than in most sectors, because the sector's own reported numbers are moving fast and the widely-circulated versions of them are frequently out of date within a quarter.

What was excluded, and why

EXCLUDED — VENDOR MATERIAL

A substantial body of published claims about AI in energy debt recovery, churn prediction and customer engagement, all of it produced by companies selling those capabilities into the sector. None traces to a regulator, ombudsman or operator disclosure. One vendor was found building sales commentary around Ofgem's own £500m debt relief figure. Excluded in full.

EXCLUDED — UNSOURCEABLE THREAT STATISTICS

Figures circulating for 2026 including "deepfakes account for 11% of global fraudulent activity" and "deepfake video scams surged 700%", which appear across fraud-prevention vendor sites without traceable methodology. Also excluded: a widely-repeated €220,000 AI voice-cloning case involving a UK energy firm, which dates from 2019, is reported second-hand, and does not name the victim.

EXCLUDED — SUPERSEDED AND UNATTRIBUTED

The figure of "£4.55bn record energy debt", repeated widely in early 2026 and already superseded: Ofgem's own data puts debt and arrears at £4.79bn as at Q1 2026. Also excluded: Citizens Advice scam figures dating from 2022, and a "43% of standard credit customers do not realise it is the most expensive way to pay" statistic that Ofgem reports as "research suggests" without attributing it.

What is flagged rather than excluded

Two positions are live and moving, and are marked wherever they appear. Ofgem's decision on its Debt Relief Scheme was expected in January 2026 and had not been published as at 4 August 2026. The EU's high-risk AI deadline is reported to have moved to December 2027; that could not be confirmed against the Official Journal, so this briefing does not rely on the date.

THE HONEST GAP

There is no credible published figure for what proportion of UK energy suppliers have deployed AI in customer-facing processes. Rather than supply one, this briefing says so.

Four findings in an earlier draft of this briefing were corrected during verification before it was written. The corrections are noted where they matter to the argument.

The market is in better shape than it has been for years

That is the honest starting point, and it makes the rest of this briefing harder rather than easier.

The domestic price cap for a typical dual-fuel household paying by direct debit stood at £1,758 for January to March 2026, down from £1,849 in April 2025 as wholesale costs eased. Prepayment customers are capped slightly lower, at an equivalent £1,711. Great Britain's electricity prices remain among the highest in Europe – 44% above the European median for households and 92% higher for medium-sized businesses – but the trajectory within the domestic market has been downward, and a further reduction was expected for the April to June 2026 period.

Twenty-two active suppliers serve the domestic gas and electricity market, with sixty-five in the business market. The six largest hold 92% of domestic share. Smart meter deployment has passed forty million installations, covering 70% of all domestic meters, with two million advanced or smart meters in the non-domestic sector.

£1,758

Price cap, dual fuel direct debit, January to March 2026

40m+

Smart meters installed – 70% of all domestic meters

82%

Domestic overall satisfaction, an all-time high

Why AI is on the agenda, stated plainly

It is not curiosity. The domestic retail sector had a financially difficult 2025: warm spring weather and rising customer debt compressed margins, and projected domestic sector profit fell to £0.27bn from £0.88bn in 2024. Non-domestic profit held broadly steady at a projected £2.47bn. On a typical domestic bill, operating costs account for 13% and debt-related costs a further 3%.

A supplier looking seriously at AI in 2026 is looking at the only substantial cost lever available that does not require a price rise or a reduction in service. That is a legitimate commercial position and it deserves to be stated rather than dressed up. The question this briefing addresses is not whether the interest is justified. It is whether the thing being bought will do what it is being bought to do.

Source: Ofgem, State of the Market Report – Energy Retail Highlights, 27 January 2026.

One further piece of context belongs here. In 2025 Tomato Energy exited the market through the Supplier of Last Resort process, affecting around 23,000 customers, and five suppliers sat below Ofgem's Capital Target at the end of September 2025, up from three at the end of June. Financial resilience in this market is a live regulatory concern, and technology spending decisions are being taken against that background.

The sector's pressure point

Every service measure is improving. Every measure of customers in difficulty is getting worse.

Domestic overall satisfaction with suppliers reached 82% in August 2025, and satisfaction with customer service reached 76%. Both are all-time highs since Ofgem began tracking them, and both now exceed pre-crisis levels. Domestic complaints have fallen to their lowest level since 2022, running at an industry average of 0.9 per 100 customer accounts in Q3 2025. The Energy Ombudsman accepted 80,256 cases in 2025, down 14% on the 92,938 it accepted in 2024.

Over the same period, domestic customer debt and arrears reached £4.79bn in Q1 2026 — up 5% on the previous quarter and 15% year on year, having already risen for twelve consecutive quarters to Q3 2025. Average arrears where there is no repayment arrangement reached £1,876 for electricity and £1,623 for gas, both up 9% on the year. Arrears without a repayment plan are now roughly three times the size of managed debt. In Q1 2018 the two were nearly equal.

WHAT THE SECTOR REPORTS

Satisfaction 82%. Customer service 76%. Complaints at their lowest since 2022. Ombudsman cases down 14%.

WHAT IS HAPPENING TO CUSTOMERS

Debt and arrears £4.79bn, up 15% on the year. Arrears without a plan now three times managed debt. 3.6m customers in debt.

The two sets of numbers are measuring different people

Ofgem's own satisfaction breakdown makes the point without needing interpretation. Overall satisfaction runs at 88% among consumers classified as "doing well", 73% among the financially vulnerable, 71% among the highly financially vulnerable, and 67% among those who report being behind on their bills. Digitally excluded consumers report 77%, social renters 76%.

The complaint numbers carry the same distortion, and here the Energy Ombudsman has said so directly. Suppliers correctly signposted consumers to the Ombudsman in an average of 48% of eligible cases in 2025, with performance ranging from 25% to 68% across the market. Ed Dodman, Chief Ombudsman for Energy, put the consequence on the record: "as many as 75% of consumers who could use Energy Ombudsman do not do so, meaning that there are potentially thousands of consumers with unresolved energy disputes."

There is also a signal inside the annual fall. The final quarter of 2025 ran 12% above the same quarter of 2024. The direction at the end of the year was upward.

THE PRESSURE POINT

The measures look best precisely where the customer is least likely to be in trouble. A model trained on this record inherits that blindness — and applies it consistently, at scale, with an audit trail.

Sources: Ofgem, State of the Market Report, 27 January 2026; Ofgem debt and arrears indicators, data to Q1 2026; Energy Ombudsman Annual Data 2025, published 20 March 2026.

What the sector is actually facing

The debt problem is one of depth, not breadth – which changes what technology can usefully do about it.

The number of customers in debt or arrears rose 5% over the past year, from 3.4 million to 3.6 million. Average debt per customer rose 11%. Ofgem's own reading of its data is that the trend "reflects a deepening intensity of debt per household, driven by rising average arrears rather than a sharp increase in the number of affected customers."

That distinction determines what a supplier should buy. If the problem were a rising tide of newly-struggling customers who had not yet been identified, better detection would be the answer. It is not. The people in difficulty have largely already been found. What has not happened is an adequate response to them.

What everyone pays

A typical consumer pays around £52 a year towards the cost of managing and writing off energy debt. Around 16% of customers pay by standard credit, which carries a £135 price premium specifically attributable to the costs of debt. Ofgem's stated aim is to drive that figure down, and it has been explicit that debt costs fall on all billpayers rather than only on those in debt.

On the causes, Ofgem has identified the home-move process as a target area: in Great Britain a consumer can generally use energy immediately on moving into a property without setting up an account, and may find themselves in debt after weeks or months. Evidence Ofgem received from suppliers suggests this cohort "may be responsible for between 20% to 40% of the overall debt figure." That is supplier-reported evidence, Ofgem reports it as such, and so does this briefing.

The repayment terms this produces

Prepayment debt repayment plans agreed during 2025 averaged 342 weeks for gas and 365 weeks for electricity across all suppliers. The variation between suppliers is extreme: the shortest average was 162 weeks for gas and 137 for electricity; the longest was 619 and 737 weeks respectively. Average weekly repayment rates ranged from £2.57 to £6.03 for gas and from £2.69 to £17.17 for electricity.

Seven hundred and thirty-seven weeks is fourteen years. It is a repayment term set by an affordability calculation, running longer than most tenancies – and in most suppliers, that calculation is already partly automated.

The human moment

A household on a smart prepayment meter runs out of credit. The meter tells the supplier. In Q1 2026, 510,393 electricity smart-prepayment customers self-disconnected at least once, and 279,430 of them for a continuous period of more than three hours. On gas, 426,505 customers self-disconnected at least once and 325,140 for more than three hours. Ofgem excludes shorter events from the headline because they "may predominantly be people that had forgotten to top up". The three-hour figure is the one that means a cold house.

Over the same quarter, no domestic customer was disconnected by a supplier for non-payment. That has been true of every quarter since Q1 2023. Disconnection has not ended in Great Britain. It has been devolved to the customer and reclassified as a meter reading.

Sources: Ofgem, Debt strategy update, 6 November 2025; Ofgem debt and arrears indicators, data to Q1 2026; Ofgem, State of the Market Report, 27 January 2026.

The challenges of using AI in this sector

The problem is not that the models are unreliable. It is that the record they would learn from has already been found unreliable, twice, by the regulator.

Ofgem has closed two major enforcement investigations into supplier treatment of prepayment customers in the last three months, and the findings in both are about the same thing.

In the OVO settlement of 3 June 2026, Ofgem concluded that the supplier "failed consistently to monitor and accurately record customer interactions, with evidence showing that key checks and safeguards were not always carried out." On the Priority Services Register, Ofgem found that "while the company had appropriate policies in place on paper, these were not reliably followed in practice", and that some customers were therefore not properly identified or supported. Staff training materials were "at times unclear, inconsistent, and contained conflicting guidance." OVO agreed to £7m into Ofgem's Voluntary Redress Fund, a £3.4m package of credit and debt relief in lieu of compensation, and £1.1m for customers in the Scottish Highlands and Islands.

In the British Gas settlement of 15 May 2026, Ofgem found that prepayment meters had been fitted under warrant where it was not appropriate to do so, in breach of licence conditions designed to protect customers in vulnerable situations. It also found that the underlying failings had been flagged to management by an external review in 2018 and an internal audit in 2021, that remedial action was inadequate, and that sufficient action was taken "only following the commencement of Ofgem's investigation." British Gas agreed to pay £20m into the Voluntary Redress Fund, to compensate customers affected between 2018 and 2021, and to write off up to £70m of energy debt for vulnerable customers under an Ofgem-agreed framework.

The regulator has published a price list for bad data

The wider Market Compliance Review, closed on 28 May 2025, required suppliers to examine their practices across more than 150,000 involuntary prepayment installations – including, importantly, smart meters that were remotely switched to prepayment mode. Ofgem found that inappropriate installation under warrant occurred in fewer than 1% of cases. Its more consequential finding was different:

"a number of customers were affected where their supplier's processes fell short including through poor data quality and inadequate record keeping, which in some cases meant customers didn't get the support they needed when they were in debt."

Ofgem then published a compensation framework, applied by eight suppliers, which prices that failure at each stage of its propagation.

DETRIMENT	COMPENSATION
Process misalignment, data quality and record keeping	£40–60
Insufficient debt support	£250
Unfair customer treatment	£250
Vulnerability not considered	£500
Inappropriate installation, switch or use of a prepayment meter	£1,000

Read downwards, that is one failure at five stages. A defect in the record is worth £40 to £60 where it is caught as a data problem. The same defect is worth £500 once it means a vulnerability was not considered, and £1,000 once it has reached the doorstep. Eight suppliers paid £5.6m in compensation to at least 40,000 customers and wrote off a further £13m of debt on top of around £55m already provided.

Sources: Ofgem enforcement settlements, 15 May and 3 June 2026; Ofgem, Suppliers commit to a further £18.6m customer compensation and debt write off following Ofgem's prepayment meter review, 28 May 2025. Organisations are named only because Ofgem published findings about them; neither has been engaged or approached.

The record you would be training on

Before any question of model quality, there is a question of what the model would be learning from.

Around 82% of domestic bills are issued against accurate readings, rising to 89% once customers supply their own. The remainder equates to approximately 54 million bills a year issued against estimated readings. Suppliers wrote off charges for at least a million customers in a year to comply with the back-billing rule at standard licence condition 21BA. Around 8% of smart meters were not operating in smart mode as at September 2025, with Ofgem taking compliance action against nine suppliers to establish why.

Four risks specific to energy retail

ONE — TRAINING DATA

The people you most need to see are missing from the data

Ofgem warns that "consumers may be digitally excluded and therefore poorly represented in digital data used, for example, in AI training sets" and that "care should be taken to ensure that any intervention does not reinforce any existing bias." Those same consumers report lower satisfaction and are less likely to reach the Ombudsman — under-represented in the input, under-served in the output.

TWO — INFERENCE

The meter reveals more than consumption

Ofgem states that "opaque AI models may enable suppliers to derive detailed insights into household energy consumption patterns or infer domestic and socio-economic characteristics that are not readily visible to consumers themselves," and that these "could influence pricing, tariff eligibility, or access to services without meaningful consumer awareness."

THREE — EXPLAINABILITY

You cannot fully test it, and you cannot fully explain it

Ofgem's own characterisation. Where a black box is used, "maloperation can occur, and be indistinguishable from normal operation without additional verification and validation." The residual uncertainty "must be considered by the stakeholder, and the organisation must decide as to whether it can tolerate the risk."

FOUR — PERSONAL LIABILITY

Competition exposure reaches named individuals

Ofgem states that "stakeholders, including IT Directors and IT Managers, can bear personal responsibility for a breach of competition law" arising from AI use, and must be able to show an audit trail confirming the system was checked for compliance. Penalties reach 10% of global turnover, with director disqualification for up to fifteen years.

Source: Ofgem, Ethical AI use in the energy sector (version 2), 13 May 2026, paragraphs 6.6, 6.20, 6.22–6.25 and Appendix 1.

AI used against you

The threat in this sector is not fraud against the supplier. It is impersonation of the redress route – and it works because of a gap the sector created itself.

It is tempting to import the threat model from financial services, where the concern is fraudulent claims and applications made against the firm. That model does not transfer here. Energy retail has a different and more specific exposure, and the evidence for it comes from the redress body itself.

On 3 February 2026 the Energy Ombudsman published a warning, still running as a site-wide banner at the time of writing, about a "recent increase in fraudulent emails being sent to energy consumers claiming to be from Energy Ombudsman". The messages tell recipients that their case has been reviewed and that they must complete a form, "often asking for personal information such as full name, address, date of birth or bank details". The Ombudsman states plainly that it will never ask consumers to open a new energy account, nor request sensitive personal information via online forms or links.

Why it works here specifically

The Ombudsman's own advice on identifying these messages is instructive. Consumers are told to look for use of its name and logo without permission, reference to a complaint they never raised, links to fake forms, pressure to act quickly, free email domains such as Gmail, and misspellings of company names.

Two of those signals – clumsy phrasing and misspelling – are exactly what generative tools remove. And the deeper problem is the third: reference to a complaint the recipient never raised only reads as suspicious to someone who knows how the real process works. Suppliers signposted consumers to the Ombudsman in an average of 48% of eligible cases. Up to 75% of consumers who could use the service never do. A consumer who has never been told the real route has no baseline against which to recognise a counterfeit of it.

THE FINDING

The sector's redress gap is the attack surface. Every percentage point of signposting failure is a percentage point of consumers who cannot tell a real Ombudsman contact from a fabricated one.

The technical threat surface

Separately, Ofgem's guidance names AI-specific vulnerabilities that suppliers should treat as novel rather than as variants of existing cyber risk: training data and model poisoning, prompt injection, and model memorisation. It notes that AI supply chains add complexity to secure system development and that, following the National Cyber Security Centre's position, a service provider "should ensure that its supply chain meets the same security standards that the organisation sets for itself."

Suppliers designated as operators of essential services carry obligations under the Network and Information Systems Regulations 2018. Ofgem is explicit that responsibility for safety, security, fairness and sustainability cannot be outsourced to an AI vendor regardless of the contractual arrangement.

Sources: Energy Ombudsman, Warning: scam emails claiming to be from Energy Ombudsman, 3 February 2026, confirmed live 4 August 2026; Energy Ombudsman Annual Data 2025; Ofgem, Ethical AI use in the energy sector (version 2), Appendices 3 and 5.

The UK legal position

There is no AI rulebook for energy retail, and Ofgem has said clearly that it does not intend to write one yet. That is the finding, not a gap.

Ofgem's stated position is that "the existing regulatory framework [is] appropriate to govern the use of AI", complemented by good practice guidance rather than replaced by new rules. Its approach is outcomes-based and proportionate rather than prescriptive, on the reasoning that prescriptive rules "would need to account for all eventualities which would be very difficult to implement effectively for rapidly developing technologies such as AI."

The consequence for a supplier is the opposite of relief. Nothing new applies. Everything old still does, and the burden of demonstrating the outcome sits with the licensee.

"There should be nothing in the guidance to excuse or prevent licensees or other regulated persons from complying with their obligations under licence conditions and other rules... Any breach of those obligations derived from the use of AI does not excuse or mitigate breach."

The obligations that actually bite

OBLIGATION	WHAT IT MEANS FOR AN AI-DRIVEN PROCESS
SLC 0 and 0A	Treating Customers Fairly. A principles-based condition, which means the outcome is judged rather than the method. An AI-driven outcome is judged the same way a human one is.
SLC 4A	Operational Capability, including the systems and processes standing behind customer decisions.
SLC 21BA	Back-billing. Already engaged by the estimated-reading problem before any model is introduced.
SLC 26, 28, 28B	Priority Services Register obligations and the safety and reasonable practicability of prepayment meters — the subject of both 2026 enforcement settlements.
NIS Regulations 2018	Security and resilience obligations for operators of essential services.
Competition Act 1998	Chapters I and II. Ofgem is a competition authority with dawn raid powers. Penalties reach 10% of global turnover and directors may be disqualified for up to fifteen years.

Ofgem's four stated AI outcomes are safety, security, fairness and environmental sustainability. Its enforcement toolkit in sectoral matters includes financial penalties of up to 10% of a regulated person's turnover, consumer redress orders, and provisional and final orders.

DIRECTION OF TRAVEL

Following the Ofgem Review of April 2026, Ofgem states that it is "changing our approach to strengthen protections for energy consumers, enhancing the focus on outcomes, and ensuring we are equipped to regulate a more complex, dynamic and innovative sector", and that it is examining how to strengthen the application of its existing statutory duties. Separately, its largest ever review of the supplier Guaranteed Standards of Performance — which have not been fully reviewed since 2015 and carry £40 automatic compensation — closed for input on 22 January 2026, with no outcome published at the time of writing.

Source: Ofgem, Ethical AI use in the energy sector (version 2), Appendix 1, 13 May 2026; Ofgem press release, 10 November 2025.

The change nobody in this sector has framed

On 5 February 2026 the rule on automated decisions about people was replaced, and energy retail has not yet worked out what that means for it.

Section 80 of the Data (Use and Access) Act 2025 came into force on 5 February 2026 by commencement regulations, replacing Article 22 of the UK GDPR with new Articles 22A to 22D. The change is significant in direction: a significant decision taken about a person based solely on automated processing moved from prohibited to permitted, provided four safeguards are in place.

- **Inform** — the person must be given information about decisions of that kind taken in relation to them
- **Represent** — they must be able to make representations about such decisions
- **Intervene** — they must be able to obtain human intervention on the part of the controller
- **Contest** — they must be able to contest the decision

A decision is "based solely on automated processing" where there is no meaningful human involvement, and the Act requires that assessment to "consider, among other things, the extent to which the decision is reached by means of profiling." Where the processing involves special category data — which includes health information held on the Priority Services Register — the tighter restrictions in Article 22B apply instead.

Why this lands harder in energy retail than the sector realises

- Ofgem's market compliance review treated smart meters remotely switched to prepayment mode as installations for enforcement purposes, alongside meters installed under warrant. A remote mode change is already regulated as an act done to a customer, not as a system configuration.
- Ofgem intends to trial processes that would "switch existing smart meters into prepayment mode in situations of domestic customers moving homes". A payment mode change triggered by a process rather than a conversation is a significant decision taken about a person at a distance.
- The Debt Relief Scheme is designed so eligible households "would be identified automatically and contacted by suppliers", using means-tested benefits data shared between DWP and suppliers. Whether the implementation meets the "solely automated" test cannot be determined, because the decision on the scheme has not been published.
- Ability-to-pay assessments are being standardised across suppliers through the scheme's second phase. A standardised assessment is, by design, one that produces the same answer from the same inputs — which is to say, one that is straightforward to automate.

THE BOARD QUESTION

Every supplier already has a route by which a customer can contest a decision. Very few could currently demonstrate that a human intervention in a model-driven decision was meaningful rather than a click-through — which is now the statutory test.

A second duty sits alongside it, unconnected to section 80's safeguards. SI 2026/425 has required the Information Commissioner to prepare a statutory code of practice on AI and automated decision-making since 12 May 2026, and it will reach every system named above, because all of them process personal data.

Sources: Data (Use and Access) Act 2025, section 80, in force 5 February 2026 by SI 2026/82 reg. 2(j), read directly; SI 2026/425, in force 12 May 2026; Ofgem, Debt strategy update, 6 November 2025; Ofgem prepayment meter market compliance review, 28 May 2025.

What is arriving next

One confirmed set of UK reforms that will reshape the complaint numbers, and one EU deadline this briefing declines to state as fact.

Confirmed — redress is getting faster, and automatic

The Department for Energy Security and Net Zero has published its response to the consultation *Fairer, faster redress in the energy market*, which ran from 23 October to 4 December 2025 and drew 42 responses. Government has confirmed it will:

- reduce the window before consumers can escalate a complaint to the Energy Ombudsman from 8 weeks to 6, with the option of reducing it further to 4
- cut the window for the Ombudsman to provide decisions from 6 weeks to 4
- ensure consumer details are automatically transferred to the Ombudsman when a case reaches deadlock with a supplier
- enable redress schemes to require compensation where suppliers fail to implement a decision, and clarify the legal standing of decisions
- bring energy brokers, price comparison services and other third-party intermediaries within Ofgem's regulatory reach, as parliamentary time allows

The automatic transfer at deadlock deserves particular attention from anyone modelling future complaint volumes. It removes the signposting step entirely — and signposting failure, at an average of 48%, is currently one of the largest single suppressors of measured complaint volume in the sector. A supplier forecasting Ombudsman case load from the 2025 downward trend is forecasting from a number that is about to be structurally reset.

Correction recorded: an earlier draft of this briefing stated the escalation window as moving from eight weeks to four. That was the proposal. The decision is eight to six, with an option of four.

Reported — an EU deadline that appears to have moved

The EU AI Act classifies as high-risk, under Annex III, AI systems intended to be used as safety components in the management and operation of critical infrastructure, including the supply of gas, heating and electricity. Customer billing and debt systems are not caught by that limb. The more relevant limb for a consumer-facing supplier would be creditworthiness evaluation — and it is worth noting that Ofgem itself describes the GB position as one in which "we effectively mandate suppliers to offer credit towards energy bills for most customers." Whether energy affordability assessment constitutes creditworthiness evaluation for Annex III purposes is a genuine open question and this briefing does not answer it.

Multiple specialist sources report that the EU's Digital Omnibus package defers Annex III high-risk obligations from 2 August 2026 to 2 December 2027, and that the Council gave final approval in June 2026. This briefing has not been able to confirm publication in the Official Journal from a primary source, and therefore does not rely on the date. For a Great Britain-only supplier the practical consequence is limited. For a group with EU operations it is not, and it should be checked against the instrument itself before any programme decision is taken on the strength of it.

Sources: DESNZ, *Fairer, faster redress in the energy market: consultation response*, 2026. EU position marked as reported and unverified — see section one.

What can be done today

Seven things, none of which requires a system that decides anything, and all of which are defensible on the evidence above.

#	OPPORTUNITY	WHY IT HOLDS
01	Fix the reading estimate	54 million estimated bills a year is the sector's largest single quality defect and the biggest driver of its largest complaint category. Forecasting consumption more accurately is a well-understood problem that transfers no accountability to a machine.
02	Restore the dark meters	Around 8% of smart meters are not in smart mode and nine suppliers are already under Ofgem compliance action. Predicting which meters will drop out, and why, is asset maintenance rather than customer decision-making.
03	Summarise the case, not the customer	Ofgem's own worked example: give the agent case history, key issues and the relevant policy faster. The agent still decides. Accountability does not move an inch.
04	Act on self-disconnection	The signal already exists in the meter data. Over 279,000 electricity customers went more than three hours without power in a single quarter. Ofgem's OVO finding was that some customers who ran out of credit received no communication or follow-up. The gap is response, not detection.
05	Instrument the record	Ofgem priced a record-keeping failure at £40–60 and its downstream consequence at £1,000. Measuring the completeness and accuracy of the customer record — and reporting it to the board monthly — is the highest-return work available in this sector.
06	Draft the AI assurance position	Ofgem's call for input on AI assurance is open and closes at 11:59pm on 12 August 2026. It asks how organisations evidence that AI systems deliver safe, fair and effective outcomes. A supplier with a written answer is in a materially different conversation to one without.
07	Map the automated decisions	Identify every point where a decision about a customer is reached without meaningful human involvement — the debt referral, the repayment calculation, the vulnerability flag, the meter mode switch — and test each against the four safeguards in section 80. This is roughly a week of work and almost nobody has done it.

THE UNIFYING POINT

Every one of these improves the record. None of them asks the record to be trusted before it has been measured.

Tomorrow, and the three levels

The useful distinction is not how capable a system is. It is where accountability sits when it is wrong.

Boards are routinely offered a capability ladder — assisted, then automated, then autonomous — that describes the technology and says nothing about who answers for the outcome. The following framing describes the same progression by accountability instead, which is the thing a board is actually being asked to approve.

LEVEL ONE

Assistive

The system prepares, summarises and surfaces. The person decides. Accountability is unchanged, and no new regulatory position is created. Almost all defensible value in UK energy retail sits here today, and every one of the seven opportunities in the previous section is at this level.

LEVEL TWO

Augmentive

The system recommends within defined bounds, showing its reasoning and a confidence level. The person confirms or overrides, and the override is recorded. The review has to be demonstrably meaningful — which is not only good practice but now also the statutory test under section 80. An override log that nobody reads is not meaningful human involvement.

LEVEL THREE

Adaptive

The system decides within a defined envelope. Oversight moves from the decision to the envelope, and accountability moves to whoever sets and monitors it. This is a reasoned projection rather than an observation: nothing in the published evidence base establishes this operating defensibly in UK energy retail today.

Three horizons

0 TO 12 MONTHS

Assistive work, plus the paperwork that makes it survivable: the register of automated decisions, the safeguard assessment, and a written AI assurance position. The unglamorous half is what survives contact with the regulator.

12 TO 36 MONTHS

Augmentive recommendation in affordability and repayment, with reasoning attached and every override logged and reviewed. Only safe once the record it draws on has been measured — which is the first horizon's work.

36 MONTHS AND BEYOND

Adaptive operation within a bounded envelope. Plausible for tariff and flexibility optimisation. Much harder to defend anywhere the output is a prepayment meter or a fourteen-year repayment term.

Honest caveat: the second and third horizons are reasoned projections, not observations. Nothing in the published evidence base establishes what UK energy retail will actually be doing in 2029, and any briefing that tells you otherwise is selling something.

Benefit realisation

If the only number that improves is the model's accuracy score, nothing has improved.

A worked example: reading estimation

Around 82% of domestic bills are issued against accurate readings, rising to 89% once customers supply their own. Billing and meter reading together generate 109 complaints per 100,000 domestic accounts — the largest single category, ahead of customer service at 88 and payment issues at 79. Back-billing write-offs under SLC 21BA affected at least a million customers in a single year.

A one percentage point improvement in first-pass reading accuracy is in the order of 540,000 fewer estimated bills a year. That is a usable business case. But it is not the benefit. The benefit is what happens downstream: movement in the billing complaint category, reduction in back-billing write-offs, and — the measure almost nobody instruments — the proportion of new debt cases that turn out to have begun with a bill the customer disputed.

What to measure, and what to refuse to measure

#	MEASURE	WHY
01	Completeness and accuracy of the customer record	The specific thing Ofgem found wanting in both 2026 settlements, and the thing every downstream model depends on. Report it to the board monthly.
02	Complaint volume and category mix	Read against your own published signposting rate. A falling complaint count with a low signposting rate is not an improvement, it is a measurement failure.
03	Outcome dispersion by cohort	Track results separately for financially vulnerable, digitally excluded and behind-on-bills customers. A model that is better on average and worse for the 67% satisfaction group has failed, whatever the aggregate says.
04	Override rate, and override review rate	On any augmentive system, measure both how often a human overrides the recommendation and whether anybody reads the overrides. The second number is the one that evidences meaningful human involvement.

THE TEST

Ask the vendor which of these four they will contract to move. The answer tells you what you are actually buying.

Sources: Ofgem, State of the Market Report, 27 January 2026; Energy Ombudsman Annual Data 2025.

A way in

Eight workstreams. The first three are a natural first step.

#	WORKSTREAM	DURATION
01	Readiness assessment	4–6 weeks
02	Use case identification and triage	3–4 weeks
03	Regulatory and legal position paper	3–4 weeks
04	Board and executive governance	4 weeks
05	Threat and integrity response	4 weeks
06	Benefit realisation framework	3 weeks, parallel
07	Build, buy or partner evaluation	4–6 weeks
08	Twelve-month mobilisation roadmap	2 weeks

The readiness assessment establishes what the customer record actually contains and where the automated decisions already are. Use case triage separates the assistive work that can start now from the augmentive work that cannot start until the record has been measured. The regulatory and legal position paper answers, in writing, the questions Ofgem is currently asking the sector and the questions section 80 now poses.

Together those three form a natural first step of six to eight weeks. Ofgem's call for input on AI assurance closes on 12 August 2026, responses are reviewed in autumn 2026, and a formal consultation on guidance design follows in early 2027 if the case is judged strong. Six to eight weeks starting now puts a supplier's position on the table before the guidance is drafted, rather than after it.

CONTACT

nc@neilcatton.com · neilcatton.com · writing.neilcatton.com

Workstream durations are indicative and scale with the size of the customer estate and the number of legacy billing platforms in play.

Sources

Every figure in this briefing traces to one of the following. All were read directly.

Regulator

- Ofgem, State of the Market Report — Energy Retail Highlights, 27 January 2026. Market structure, price cap, satisfaction, complaints, smart metering, supplier financial position, billing accuracy.
- Ofgem, Debt and arrears indicators, interactive data, latest quarter Q1 2026. Debt totals, arrears, repayment terms, self-disconnection, disconnections for non-payment.
- Ofgem, Debt strategy update: supporting the reduction of energy debt, 6 November 2025. Debt cost per bill, standard credit premium, home moves, Debt Relief Scheme design, involuntary prepayment evaluation.
- Ofgem, Ethical AI use in the energy sector (version 2), 13 May 2026. Ofgem's four AI outcomes, governance and risk practices, sector-specific examples, transparency and explainability, legal and regulatory obligations.
- Ofgem, Call for input: AI assurance in the energy sector, 17 June 2026, closing 12 August 2026. Deployment areas, assurance lifecycle, consumer protection and fairness questions.
- Ofgem, Debt Relief Scheme: statutory consultation, published 6 November 2025, closed 19 December 2025, status "closed (awaiting decision)" as at 4 August 2026.
- Ofgem press release, British Gas settlement, 15 May 2026.
- Ofgem press release, OVO settlement, 3 June 2026.
- Ofgem press release, Suppliers commit to a further £18.6m customer compensation and debt write off following Ofgem's prepayment meter review, 28 May 2025. Compensation framework and market compliance review findings.
- Ofgem press release, Ofgem unveils plans for overhaul of energy customer service standards, 10 November 2025. Consumer Outcomes and Guaranteed Standards calls for input.

Ombudsman

- Energy Ombudsman, Annual Data 2025, published 20 March 2026. Case volumes, category mix, signposting rates, Chief Ombudsman statement.
- Energy Ombudsman, Warning: scam emails claiming to be from Energy Ombudsman, 3 February 2026, confirmed live 4 August 2026.

Legislation and government

- Data (Use and Access) Act 2025, section 80, read at legislation.gov.uk. In force 5 February 2026 by SI 2026/82 regulation 2(j). New UK GDPR Articles 22A to 22D.
- DESNZ, Fairer, faster redress in the energy market: consultation response, 2026. Consultation ran 23 October to 4 December 2025 and received 42 responses.

Marked as reported, not verified

- EU Digital Omnibus deferral of AI Act Annex III obligations from 2 August 2026 to 2 December 2027. Reported by multiple specialist and legal sources; not confirmed against the Official Journal. Not relied upon in this briefing.

About the author

Neil Catton — Independent Technology Strategist

Thirty-eight years across more than twenty sectors, from computer operator to Group CTO. CTO and director-level roles at BT, Wipro, Fujitsu, Sopra Steria and Capita. Author of the technology ethics trilogy *The Next Evolution*, *The Cognitive Crucible* and *The Shadow System*, and of the standalone *Working as Designed*. He writes regularly at writing.neilcatton.com on the intersection of technology, human consequence and organisational systems.

The career began at British Nuclear Fuels and has returned to energy, utilities and critical national infrastructure repeatedly since — most substantially as one of three architecture directors on BT's £1.5bn transformation programme, with £500m of allocated budget spanning digital, data, integration, cloud and AI across a hundred-strong architecture practice. The regulated-consumer half of this briefing draws on work in banking and insurance, including two challenger bank builds taken through the financial services licensing process, where the question that runs underneath this document — can you evidence that a decision about a customer was made properly — has been asked for longer, and answered harder.

The practice is a fractional and advisory portfolio spanning public safety, public sector, insurance, health technology and early-stage ventures: board-level technology strategy, independent challenge, and the unglamorous work of making an organisation able to explain itself.

How this briefing was produced

Research was restricted to regulators, ombudsmen, published legislation, government consultation outcomes and operators' own disclosures. Every primary document behind a load-bearing figure was read in full rather than summarised from search results, and the publication date of every page was checked. Where a figure could not be traced to a qualifying source it was excluded and the exclusion recorded in section one; six figures were excluded on that basis. A verification pass before writing corrected four findings, including the status of Ofgem's Debt Relief Scheme, the confirmed escalation window in the DESNZ redress reforms, and the current title of Ofgem's chief executive.

NOTICE

1. **Not advice.** This briefing is not legal, regulatory, financial or professional advice and should not be relied on as such.
2. **Position as at a date.** The position is stated as at 4 August 2026. Ofgem's Debt Relief Scheme decision, its evaluation of the involuntary prepayment meter rules, the outcome of its Consumer Outcomes and Guaranteed Standards calls for input, and the EU high-risk AI timetable may all have moved since.
3. **Third-party sources.** Sources are cited in good faith. Verify before relying on them.
4. **Named organisations.** Organisations appear only where a regulator or ombudsman has published a finding, and are used illustratively. None has been engaged or approached in connection with this briefing.
5. **No relationship created.** No advisory or client relationship is created by receipt of this document.
6. **Independence.** The author holds no vendor affiliations, reseller agreements, referral arrangements, commission interests or equity positions in any product, platform or supplier referenced in this briefing, and has received no payment from any organisation named in it. This is what makes the exclusion discipline in section one credible rather than merely stated.
7. **Copyright.** © Neil Catton 2026. May be circulated internally within the recipient organisation as required.